

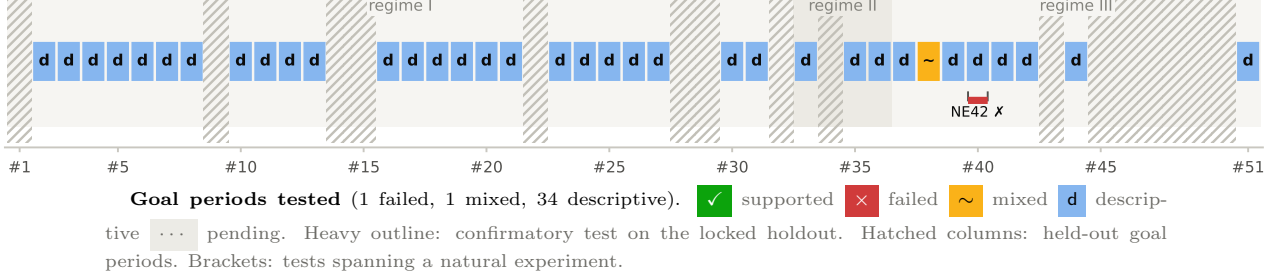
H85: Scaling of outputs with N

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Talk is a shared, nearly fixed channel, not a per-agent output: messages scale as $N^{0.33}$ while addressing per message rises as dilution predicts.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

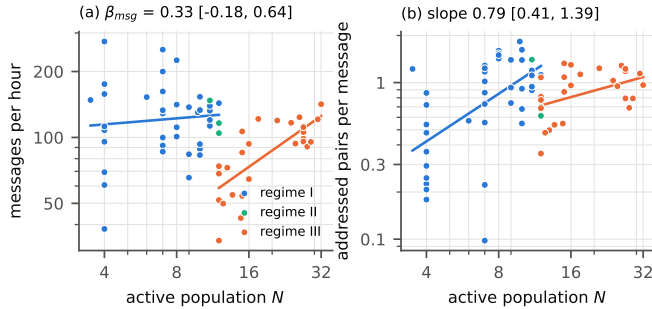
The test asks how the swarm's outputs scale with its active population N . Each of 71 non-holdout period units gives one point: output per scheduled hour against N , with regime intercepts and goal-cluster bootstrap CIs. Talk is sublinear: messages scale as $N^{0.33}$ $[-0.18, 0.64]$, and in regime I the room's talk volume is nearly fixed ($\beta = 0.09$). Addressing per message rises as $N^{0.79}$ $[0.41, 1.39]$, against 0.44 predicted from H18's dilution and the measured pending set ($k \propto N^{1.29}$). Reply parents per message do not rise (0.02), so the HH's reply kill fires. Work exponents are not identified. The holdout script exists and is not run.

PHYSICAL MODEL

Allometric scaling (model 14) with a per-call response kernel (model 02). Y_u is a unit's total output, T_u its scheduled hours, N_u its hour-weighted active population. α_r is a regime intercept and ζ_g a goal-period effect. Responses per talk call scale as $k^{1-\beta_d}$ with pending set $k \propto N^{\gamma_k}$ and H18's $\beta_d = 0.66$. DQ2 keeps one reply parent per message, which caps reply counts.

$$\ln(Y_u/T_u) = \alpha_{r(u)} + \beta_Y \ln N_u + \zeta_{g(u)} + e_u$$

$$\beta_{\text{ment}} - \beta_{\text{msg}} = (1 - \beta_d) \gamma_k, \quad \beta_{\text{reply}} - \beta_{\text{msg}} \approx (1 - \beta_d) \gamma_k \frac{\lambda(1-f)}{f}, \quad f = 1 - e^{-\lambda}$$



Observables. (a) Messages per scheduled hour against active population, one point per unit, colored by regime, with within-regime fits. Regime I is flat: more agents do not add talk. (b) Addressed agents per message against N . It rises in every regime, steepest in regime I.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not expect more talk from more agents. In regime I each agent kept its call clock, but its share of talking calls fell as $N^{-0.89}$, so the room's message rate stayed near 130 per hour (IQR 96–148) from 4 to 11 agents. In regime III talk grows as $N^{0.77}$. A message in a larger room names more agents: addressing per message grows as $N^{0.5}$ to $N^{0.8}$. Expect it, and route long rosters into rooms: within a day, the larger room talks less per agent (room exponent 0.36 in #38). Do not read reply-graph density as interaction load: reply parents per message stay flat in N (median 0.35 in regime I, 0.51 in III). Budget ± 0.5 on any output exponent fitted across goal periods.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 messages $\beta \in [0.85, 1.15]$	0.33 $[-0.18, 0.64]$; I 0.09, III 0.77	fail
P2 replies $\beta \in [1.15, 1.55]$	0.36 $[-0.46, 0.87]$; per message 0.02 vs 0.32	kill fires
P3 addressing/msg $= 0.34\gamma_k$	0.79 $[0.41, 1.39]$ vs 0.44 $[0.31, 0.52]$	pass
P4 $\gamma_k \in [0.7, 1.3]$	1.29 $[0.91, 1.52]$; rooms 1.02 $[0.88, 1.15]$	pass
P5 work $\beta = 1.0 \pm 0.1$	1.89 $[0.55, 7.76]$ (36 units)	unident.
P6 repos own vs shared	CIs span > 5	unident.
N1 NE42 merge	placebo $\times 2.5\text{--}7.5$; k ratio 1.34	fail
N2 #51 roster sweep	SD $\ln n$ 0.08; CIs span > 2	unident.
N3 #38 room contrast	msg 0.36 $[0.02, 0.82]$; ment 1.12	mixed

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 d descriptive
+26 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

One decade of N , 35 goal clusters and regime intercepts give a level-exponent SD of 0.25 (synthetic), not the design 0.10. N grew with calendar time, so model generations ride along. Regime-I chat mode showed agents a capped recent-chat window, which may shape the talk budget. Addressed pairs count names, including co-addressing (post hoc: names per naming message rise as $N^{0.44}$ in regime I). Reply parents carry DQ2's one-parent budget. Room contrasts have no room fixed effect. NE42 and the #51 sweep could not identify a step.

ROUND 2

Fit a talk-budget model per call (talk probability against room traffic). Restrict addressing to senders with a pending message (the clean aggregate of H18). Measure work per incumbent agent across roster joins with agent fixed effects. Run the frozen holdout test after sign-off.

- **Essence.** whether the swarm produces more per agent as it grows (a collective benefit), and how much of its talk is a shared, capacity-limited channel.
- **H85-R1** Talk budget model.
- **H85-R2** Pending-sender addressing.
- **H85-R3** Work per agent across joins.
- **H85-R4** Uncapped replies.

Second figure. (a) Exponents across 71 units (blue, 95% goal-cluster CIs) against predictions (orange): the reply prediction is 1.25 after the one-parent budget. (b) Within-day room-size exponents in the four rooms-era periods where room sizes vary, with the random-effects mean (squares).

